Beyond the classic 16S rRNA classification: revealing two distinct species within *Methanobrevibacter smithii* through comparative genomics

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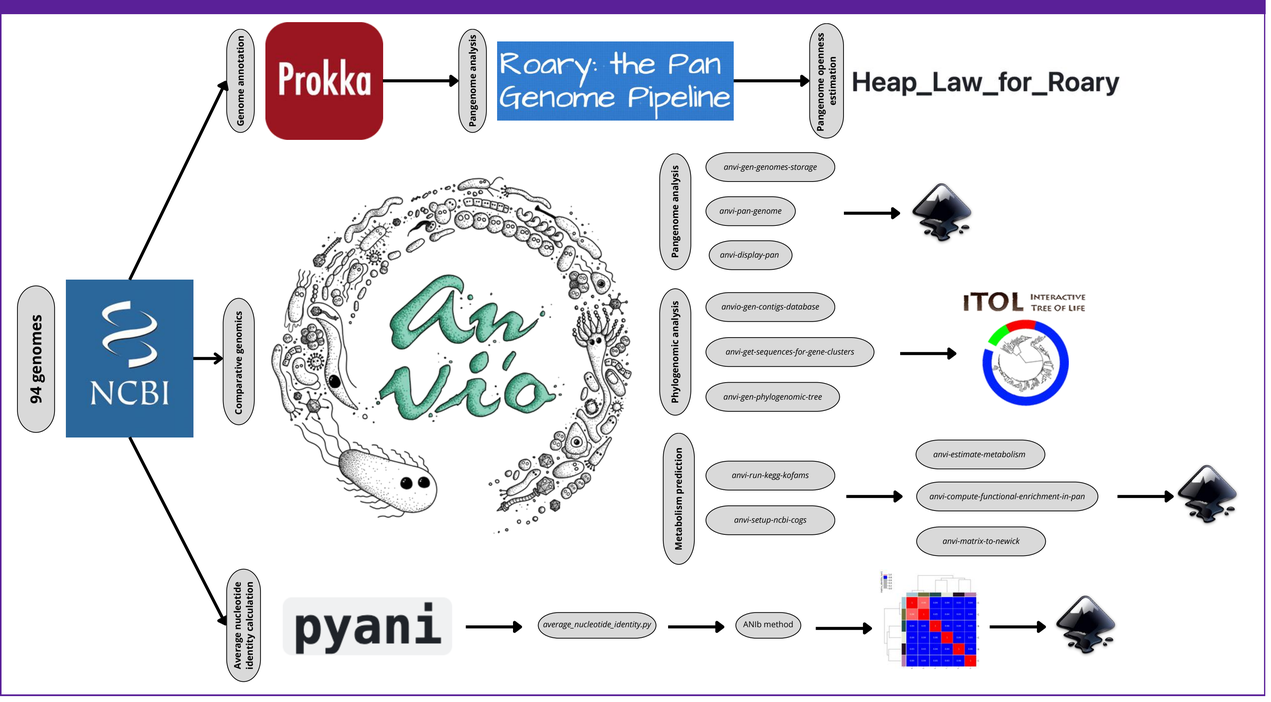
Introduction

Archaea are present in various parts of the body, including the intestines, skin, vagina, lungs, and nasal and oral cavities (1). The methanogenic species *Methanobrevibacter smithii* represents between 1% and 10% of the intestinal microbial community, highlighting its ubiquitous presence and potential impact on human health and disease (2). Although the 16S rRNA gene has traditionally served as the foundation for microbial classification, its effectiveness in distinguishing closely related species is limited, as exemplified by *M. smithii*, which actually comprises two distinct species (2,3). Therefore, it is essential to utilize tools beyond 16S rRNA to resolve taxonomic ambiguities and accurately define species-level differences.

Materials and methods



Methanobrevibacter smithii have an open pangenome

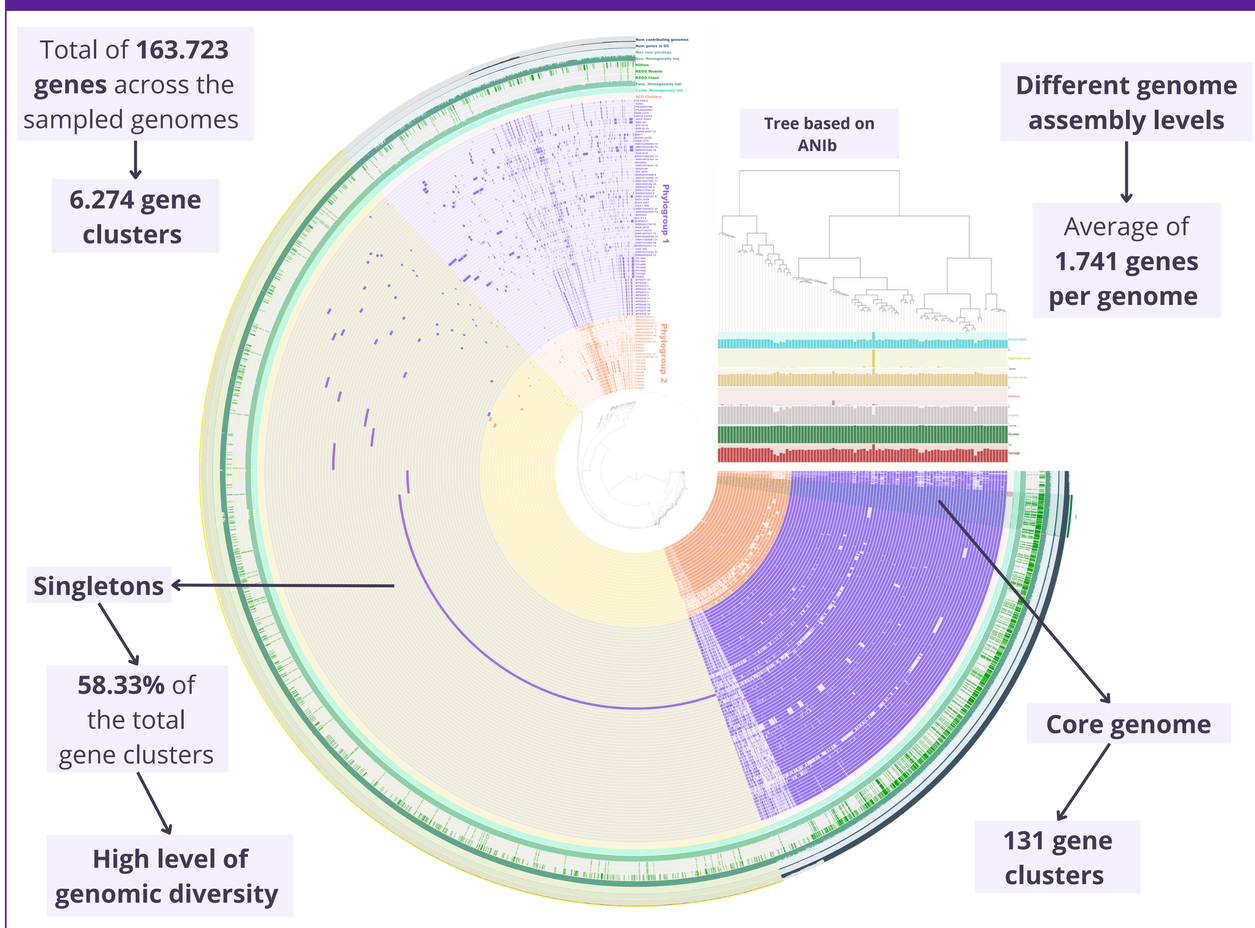


Figure. 2. Graphical representation of *Methanobrevibacter smithii* pangenome. This visual representation organizes genomes based on a gene presence/absence clustering tree and distinguishes the two phylogroups with unique color schemes, lavender and peach color for phylogroup 1 and 2, respectively. Outer rings detail single-copy gene clusters, gene counts, contributing genomes, and homogeneity indices, with core and singleton genes highlighted in green and yellow, respectively. Key genomic metrics are displayed as bar charts, and the top-right features a ANIb-based phylogenomic tree. KEGG modules with known functions were denoted in green and unknown functions in gray.

Results and discussion

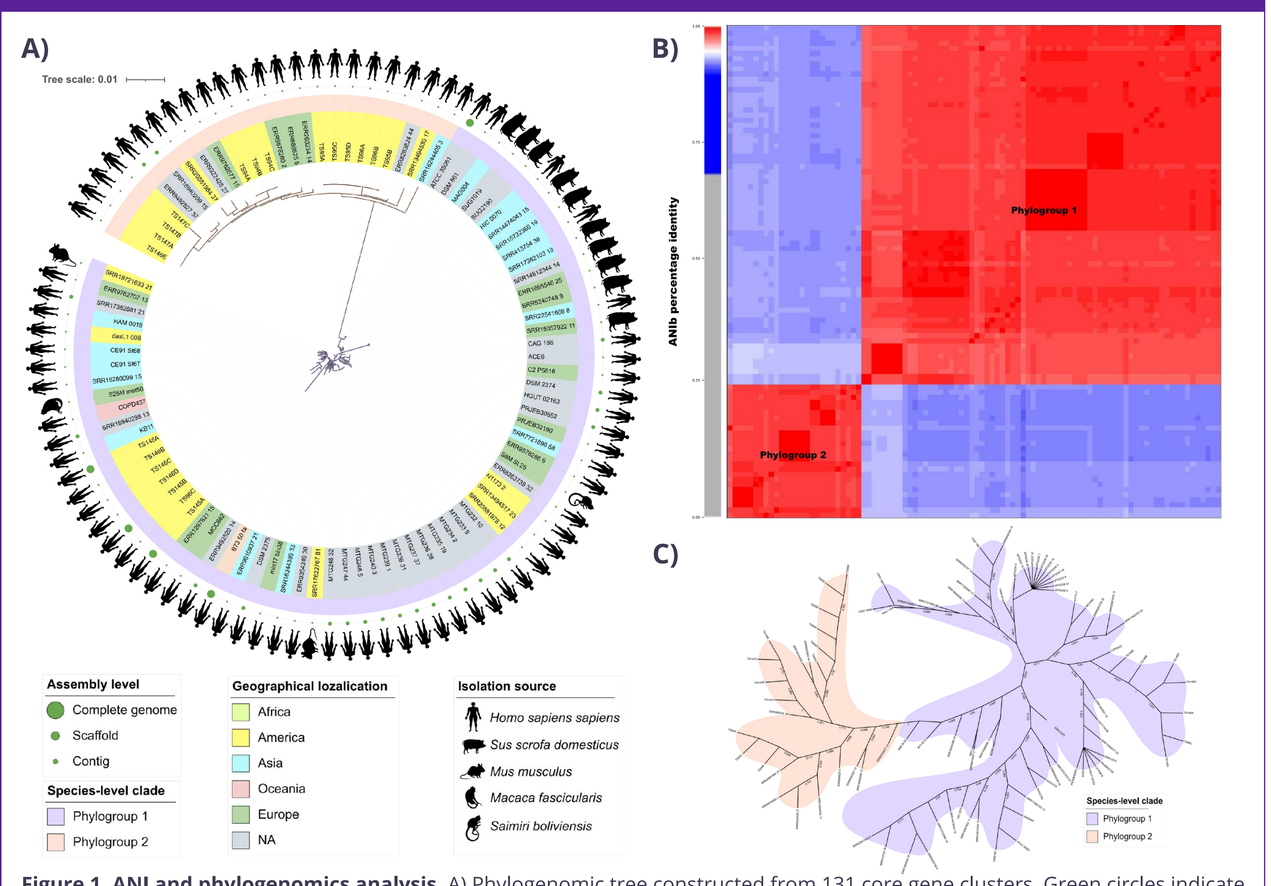


Figure 1. ANI and phylogenomics analysis. A) Phylogenomic tree constructed from 131 core gene clusters. Green circles indicate genome assembly level, while geographical distribution is color-coded, and silhouettes around the tree represent the diverse host species from which the strains were isolated. B) Average Nucleotide Identity (ANI) heatmap of 94 genomes, organized into two phylogroups. Each cell represents the ANI value between genome pairs, with darker red indicating higher similarity (≥98%) and blue suggesting lower similarity (≈93%). C) Phylogenomic tree based on 76 single-copy core proteins from the same 94 genomes. Phylogroups are consistently represented across all figures, with phylogroup 1 corresponding to a peach color and phylogroup 2 to a lavender color.

Comparative pangenome analysis and metabolism prediction

Pathogenicity

- ***Soj***: involved in the sporulation of *Bacillus subtilis*
- ***CapA***: associated with capsule formation
- ***apxIB***: involved in exotoxin production
- ***SrfB***: virulence factor with unknown function
- ***RhuM***: uncharacterized *Salmonella* virulence factor

Antibiotic resistance

- ***MepA***: implicated in multidrug resistance
- ***aacA-aphD***: resistance to aminoglycosides
- ***sttH***: streptothricin resistance
- ***tetA***: tetracycline resistance
- ***bcrA-B***: bacitracin resistance

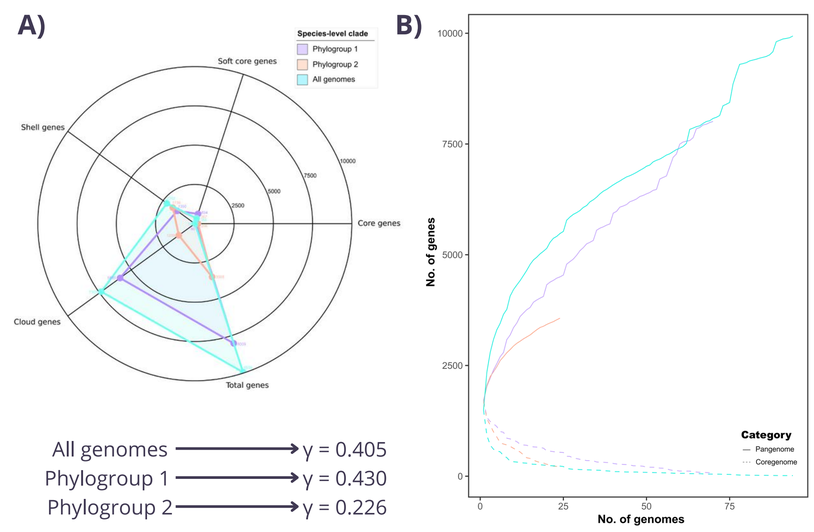


Figure 3. Comparative pangenome approach. (A) Radar chart showing pangenomic composition, distinguishing between core, shell, soft core, and cloud genes across all genomes combined, as well as for each of the two phylogroups. (B) Pangenome accumulation curves for all genomes collectively, and for each individual phylogroup.

Conclusions

The phylogenomic and ANI analyses, combined with functional profiles and genomic divergence, support the reclassification of *Methanobrevibacter smithii* into *M. smithii* and *M. intestini.* The identified genes related to antibiotic resistance and pathogenicity may be crucial for understanding these species' roles in the human microbiota and could challenge the view of archaea as non-pathogenic. However, further research is needed to fully explore this possibility. This case highlights broader issues of taxonomic misclassification in prokaryotes and underscores the importance of advanced genomic approaches for achieving accurate microbial classification.

References

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3. **Low A, Lee JKY , Gounot JS, Ravikrishnan A, Ding Y, Saw WY, et al.** Mutual Exclusion of *Methanobrevibacter* Species in the Human Gut Microbiota Facilitates Directed Cultivation of a *Candidatus* Methanobrevibacter Intestini Representative. Xu ZZ, editor. Microbiol Spectr [Internet]. 2022 Aug 31;10(4). Available from: https://journals.asm.org/doi/10.1128/spectrum.00849-22